

CENTURY NIT CONSULT

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# Complete Platform Documentation

The study-abroad platform end to end. What it does, how it is built, how the client journey works, and how it runs in production.

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*Everything about the platform in one document, in plain language. It covers what the system is, how it's built, what each part does, how the client journey works end to end, and how it runs in production.*

# What the platform is

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Century NIT is a study-abroad consultancy platform. A client signs up, books a paid consultation, and is then guided step by step, by real staff, through enrolment, university applications, visa processing, and departure. Staff run the whole operation from a separate console.

The product is **three deployable pieces**:

1. **The public website and client portal**, where visitors browse and where clients live their journey.
2. **The Operations Center**, the staff-only console for consultations, cases, finance, marketing, content, and administration.
3. **The API**, the single backend that both front ends talk to. It holds all data and all business rules. It also publishes an interactive reference (every endpoint, its inputs, and a working "try it" button) for anyone integrating against it.

A visitor to the website never downloads a byte of the staff console. They are separate applications with separate builds.

## HOW THE PIECES CONNECT

Each front end sits behind its own Cloudflare Worker. The Worker serves the app's files and forwards every API call to the backend. Because of that forwarding, the browser sees one origin per app, the login cookie is first-party, and sign-in just works. The API keeps a strict list of allowed origins for the rare calls that are genuinely cross-origin; the same list feeds both the browser security layer and the sign-in system so they can never drift apart.

## WHERE THE TRUTH LIVES

Everything of record is in one PostgreSQL database: cases, invoices, documents, messages, marketing, content. Both front ends are thin clients. They show what the API says and send back what the user does. The portal keeps a little per-browser convenience state (drafts, dismissed hints) and an offline fallback for the journey view, but nothing there is authoritative.

## PART 2

# Signing in

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One identity system serves both audiences. A staff member is simply a user account that also has a staff record attached; that record is where their role and branch live.

## CLIENTS

Clients register themselves. They can sign in with:

- Email and password (with email verification and password reset)
- Google
- Phone number with a texted code (only when a text provider is configured; without one the option refuses rather than pretending)
- A one-time code emailed to them (passwordless)

Clients can add two-factor protection, an authenticator app or emailed codes, but it's never required. An applicant is not made to install an authenticator before they can book a consultation.

## STAFF

**Staff can never register themselves.** There is no staff sign-up anywhere. An account exists only because someone senior invited it, and the invitee sets their own password; nobody else ever knows it. Invitations and one-time codes go out by email; in development they print to the server console instead so everything can be tested without email accounts.

Two-factor is **required for every staff role**, enforced by the server: either an authenticator app or enrolled email codes. The very first administrator is created once with a server-side setup token; the moment any staff member exists, that door closes permanently.

## PART 3

# Who can do what

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Two separate ideas, both checked on the server (the console interface only hides buttons; the backend is the real authority):

- **Modules** decide what a role can *see*: which pages and API areas open to them.
- **Capabilities** decide what a role can *do*: assign work, issue invoices, invite staff, own a chapter of a case.

Both live in one shared definition used by the console and the API, so they can never disagree.

Administrators can build custom roles in a permission matrix; a custom role gets a rank, and you can only grant roles ranked below your own.

## THE BUILT-IN ROLES

| ROLE             | SEES                                                                                            | CAN DO                                                                                                                            |
|------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Super Admin      | Everything                                                                                      | Everything                                                                                                                        |
| Admin            | Platform administration (staff, sign-in policy, content, settings) plus read access to casework | Invite and manage staff, manage roles and client accounts, change platform settings; does not work case files                     |
| Manager          | All branches, all cases                                                                         | Assign work, invite staff, issue invoices, approve payment schedules, own any chapter, edit packages and the university catalogue |
| Coordinator      | All cases and branches                                                                          | Assign and route work, own any chapter; the front-desk role                                                                       |
| Customer Service | Cases at their branch, leads, helpdesk                                                          | Route work, first-line support                                                                                                    |
| Consultant       | Their own caseload                                                                              | Own chapters (consultations, applications, visa, departure), comment, request documents, update stages, raise draft invoices      |
| Finance          | Invoices, ledger, payments, packages                                                            | Issue, void and credit invoices, edit packages, approve payment schedules                                                         |

Branch scoping is real: a role without "see every branch" is confined to its own branch, enforced by the API.

## PART 4

# The client journey

Everything hangs off one ladder: **six chapters**.

| CHAPTER                | WHAT HAPPENS                                                                                                                        |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1: <b>Consultation</b> | Client books online or in-person, picks a real slot, pays, meets the consultant, gets an eligibility assessment                     |
| 2: <b>Enrolment</b>    | Client confirms they want to proceed (or pauses or declines), picks a service package and payment plan, pays the 10% agency deposit |
| 3: <b>Applications</b> | Client chooses schools, pays the application fee, staff submit and track offers, client accepts one                                 |
| 4: <b>Visa</b>         | Client consents to start, pays the visa fee, the officer files and tracks it to a decision                                          |
| 5: <b>Departure</b>    | Travel help (yes, hold or no), the pre-departure fee milestone, a checklist, then official documents released                       |
| 6: <b>Complete</b>     | Client departed; post-arrival instalments run as aftercare                                                                          |

Inside the chapters the portal walks the client through about fourteen finer steps. A step is never silently skipped: if later evidence exists (an admission, a paid visa), earlier un-ticked steps show as "skipped", not done.

## HOW "WHERE THE CLIENT IS" IS DECIDED

Two representations, deliberately different:

- **A coarse stage stored on the case**, set by staff and checked against gate rules (you can't enter visa work before the fees settle). It acts as a floor: it can push a client forward, never backward.
- **A fine step computed from facts**. The API gathers what actually exists (a booking, a paid deposit, an offer, a consent) and derives the step the portal shows. It is never stored, so it can't disagree with reality, and both apps use the same shared logic and the same words.

Everything is scoped to the client's **current case**. A returning client's earlier case can never leak paid invoices or admissions into the new one; schools and invoices are read per case, never per person.

## CONSENT IS A GATE, NOT A SUGGESTION

The client's Confirmed / On hold / Declined is recorded and gates stage entry. Staff cannot push a client past a consent they haven't given. The visa chapter in particular opens only on explicit consent.

## WHO OWNS THE WORK

Ownership history is append-only. Who owns a case now is the open record; reassignment closes the old one, never overwrites. Each chapter has its own seat (consultant, visa officer, travel officer), and the assignment picker for each chapter only offers staff holding the matching ownership permission. A consultant can delegate a session to a colleague and reclaim it, and a daily duty-coordinator roster covers each branch.

## HARD BOUNDARIES

Some moments stop the case until a manager acts. Paying the deposit raises a "pending handler assignment" handoff; that payment is the only trigger for the school-submission handler. Entering departure or the final payment stage hard-gates the same way until a manager resolves the handoff or defers it.

## FLEXIBLE ENTRY AND EXIT

Service packages are priced per stage. A client can enter at different service stages, select only what they need, stop after any reached stage, and later request to continue. A continuation is a real flow, not just a flag: staff propose the next stage and attach a quotation for it, the client accepts, pauses or declines from the portal, and a lapsed or declined offer can be re-invited. Accepting opens the next chapter under its own pricing, so a client who stopped after applications can resume at visa without buying a new package.

## WHO DOES WHAT, WHEN

| MOMENT          | ACTOR                         | ACTION                                                                       | CLIENT SEES                                 |
|-----------------|-------------------------------|------------------------------------------------------------------------------|---------------------------------------------|
| Booking lands   | Coordinator or manager        | Assign a consultant (or the duty roster covers it)                           | "Consultant being assigned"                 |
| Session held    | Consultant                    | Complete the assessment (eligible / conditional / not eligible, with a note) | Assessment outcome                          |
| Enrolment open  | Client                        | Confirm, hold or decline; pick package and plan; pay the deposit             | Journey advances                            |
| Deposit settled | Manager                       | Resolve the handler handoff, assign the case consultant                      | Named consultant                            |
| Schools chosen  | Chapter owner, then finance   | Raise the draft invoice, approve and issue it                                | "Fee being prepared" then a payable invoice |
| Offers in       | Consultant                    | Record each school's outcome                                                 | Live tracking; client accepts an offer      |
| Visa start      | Client, visa officer, finance | Client consents; officer raises the draft, finance issues, client pays       | "Pay the visa fee"                          |
| Visa work       | Visa officer                  | Embassy filing, biometrics, decision                                         | Visa tracking                               |

|                    |                    |                                                |                              |
|--------------------|--------------------|------------------------------------------------|------------------------------|
| Departure boundary | Manager            | Resolve the handoff, assign the travel officer | Departure chapter opens      |
| Travel settled     | Client and finance | Flight choice; fee milestone paid              | Documents on the unlock path |
| Pre-departure      | Client and staff   | Checklist items verified or waived             | Official documents released  |
| Any time           | Finance            | Issue, void, credit invoices; edit packages    | Ledger and receipts          |
| Any time           | Admin              | Staff invites, roles, sign-in policy, content  | (No client-visible step)     |

## Consultations, scheduling and meetings

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What a client can book comes from three inputs: the branch's weekly hours and slot settings, each consultant's own working hours, and everything already booked, including busy times mirrored in from staff personal calendars. A database-level rule makes double-booking impossible even if two people click at the same instant.

Bookings have a full lifecycle (cancel, reschedule, reassign, confirm, complete, no-show, rebook credit), and every transition is written to an append-only history. Reschedules a client requests go to staff for approval; staff-side reschedules are direct.

Online sessions run in video rooms the client joins **inside the portal**, no external link needed. A join window is enforced (opens a few minutes early, closes after). A background check every minute marks meetings live or ended, feeding the console's "what's happening now" view.

Staff calendars connect both ways: they can pull their personal calendar's busy times in so clients can't book over them, and publish their Century bookings out to their personal calendar as a subscribeable feed.

# Money

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All amounts are stored as integer cents. The invoice pipeline is a two-step approval chain that drives the journey:

1. The **chapter owner** (the consultant or visa officer doing the work) raises a draft estimate, a proforma, on the case.
2. **Finance** approves and issues it. Now it's a real, payable invoice.
3. The client pays; partial payments count; when the balance reaches zero the invoice settles.
4. Corrections happen by voiding or crediting. Every action lands in the invoice's permanent event history, never by editing the past.

## HOW CLIENTS PAY

Paystack is the primary client-side rail: checkout, then verification, then a signed webhook settles the payment on the server. The webhook is the authority; if it's slow, staff can reconcile against the gateway. A Stripe card gateway exists alongside it and activates when a key is configured, so the payments log shows records from both gateways plus staff-recorded bank and cash entries. There is deliberately no way for a client to claim "I paid"; cash and bank payments are recorded by staff. Mobile-money charges with an OTP step are also supported. Receipts and invoices print as PDFs.

## INSTALMENTS

Clients can pay in instalments: full payment or a plan at enrolment, and a post-arrival schedule they propose and a manager or finance approves. With the client's consent, a saved payment authorization lets the system charge each instalment automatically; every attempt is logged so failures are visible, never silent.

## FEES AND PACKAGES

The fee catalogue defines what each charge is (agency service fees, plus third-party fees like embassy and school application charges), with tariffs per destination. Service packages are priced per stage so clients buy only the chapters they need. Charges are priced in dollars and charged in cedis: a configurable exchange rate converts them, so the amount a client actually pays tracks a rate finance controls. A post-arrival catalogue carries the charges that only become relevant after the client lands, each with its own trigger and instalment rules.

## THE LEDGER

Every case and every client has a ledger view: a chapter-numbered journal of invoices and payments. Underneath sits the invoice event history, immutable and append-only, the substrate the views are built from. A paid invoice can't be voided (it must be credited), and settlement is idempotent, so a replayed webhook or

retried verification can never charge twice.

## Documents

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Clients upload through a vault: the file goes straight to private storage via a temporary signed link, then staff verify it or reject it with a note. Verification progress feeds the client's checklist and the journey on both sides. Staff can request a specific document by name ("passport data page"), which appears as a task in the portal. Rejected files are purged after a retention period. Nothing is publicly reachable; every download goes through a short-lived signed link the API mints.

One deliberate hold: the documents the agency produces as the client's agent, the admission letter and the visa outcome, are visible in the vault but locked from download until the pre-departure fee milestone is paid. A manager can release them early, and that release records a reason on the case. Staff downloads are never held; the lock is a client-side gate only.

# Communication

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## ONE CONVERSATION SYSTEM, THREE SURFACES

A single thread model (conversations, members, messages, attachments, mentions, reactions) serves:

- **The helpdesk:** client-to-staff threads, scoped to a case and chapter, with subject, category, priority, who it's waiting on, first-response and resolution times, and a satisfaction rating when it closes.
- **The staff Communication Hub:** direct messages, group chats, mentions, reactions, attachments, presence, typing indicators.
- **The client's Communication Center:** the portal's thread with their assigned staff plus the AI assistant.

Threads can be escalated to managers, and staff keep a library of canned replies. When a client is offline, a reply is emailed to them, and their email reply lands back on the same conversation. Staff presence and typing are shown live as they happen.

## THE AI ASSISTANT

A knowledge assistant runs at the edge on Cloudflare's Workers AI (a Llama 3.1 model, streaming its replies as they're generated), with three personas: a **website assistant** for prospective students on public pages, a **portal assistant** inside the client's communication hub, and an **enquiry** path for visitors. It answers general questions only, says so when it isn't certain of a Century-specific detail, and points people to a human for anything account-specific. First-time public use is gated by a bot check so the widget can't be farmed.

## NOTIFICATIONS

One notification service fans out to three channels (in-app, email, and browser push) and logs every delivery attempt so a failure can be retried. Clients only ever see client-appropriate event kinds; staff-only signals never reach the portal. Each user controls their own matrix of which events arrive on which channels. Email never sends inside a web request; it's queued, so a failed send can never roll back a booking.

The portal is push-driven: a live event stream tells it when stages, assignments, invoices or visa states change, so the client sees updates without refreshing. Polling exists only as a fallback for a dropped stream.

## THE AUDIT TRAIL

Administrative actions land in an append-only audit trail: who, what, on which record, from which address. Every entry seals the previous entry's fingerprint, so altering or deleting any row breaks the chain and a verification check proves it. The trail is browsable by day, filterable, exportable, and alert rules can watch it (for example, a burst of failed sign-ins or a role change). Settings changes have their own parallel trail.

# Marketing

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Production email marketing built on the suite's own data, in five console tabs: Campaigns, Audiences, Contacts, Templates, Automations.

## PEOPLE, NOT ADDRESS BOOKS

The **person** is the record; a list is a membership. The contacts view gathers everyone the suite knows (mailing-list subscribers, applicants, leads), each with their real identity (an applicant shows their case and chapter, a lead their stage), their consent state, and their memberships. One person with two memberships appears once, never as two rows.

## CONSENT IS EARNED, NEVER FABRICATED

Adding a contact has exactly two doors: send them a confirmation email (they become opted-in when they click), or record that they consented offline, which requires a written note as the audit trail. There is no silent "confirmed". CSV imports run a dry run first (how many are new, how many duplicates merged, how many invalid), then apply the same consent decision to the batch.

## SUPPRESSION

One global do-not-contact list, checked twice: when a campaign's audience is frozen (only opted-in, unsuppressed people become recipients; everyone else is recorded as skipped with the reason) and again at the instant each email is sent. Someone who unsubscribes between scheduling and sending is skipped, not mailed. Bounces and spam complaints from the email provider suppress automatically.

## SEGMENTS: LIVE AUDIENCES

Saved filters over the suite's real data (applicants by chapter, branch, country, offer state, unpaid milestone, departure window; leads by stage, source, no-show), evaluated at send time, never copied. A preview shows how many match, how many are opted in, never asked, or suppressed. A campaign targets a list or a segment, never both.

## CAMPAIGNS

Composed from blocks (headings, paragraphs, buttons, dividers, two columns) with merge fields that pull real data: the person's name, case reference, stage, officer, branch, next due date, portal link. Senders get a preheader, a from-name, and a reply-to (the branch mailbox by default). Preview is the actual server-rendered email, and a test send goes through the real pipeline. Reports show delivered, open, click, bounce and unsubscribe rates, an hourly timeline, top links, and the per-recipient ledger with retry for failures. Every footer carries working unsubscribe and preferences links.

## TEMPLATES AND AUTOMATIONS

Templates render as live miniatures; presets are read-only starters you fork. Automations are rules (when this happens, to people matching this segment, send this template after this delay) covering moments like a no-show follow-up, an assessment completed but no enrolment, an offer received, a visa approval, thirty days to departure, or an overdue milestone. Starters ship as drafts; staff turn them on deliberately. Every firing is logged, including why a person was skipped.

## Content management

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One identity record, one content store, one media library, read by the website, the portal, the console, and the email layout.

- **Brand:** the identity record (names, logos, colours, contacts, socials) has a draft and a published copy. Edits save to the draft; publishing makes it live and records who published, with full version history and revert.
- **Entries:** content collections keyed by name and slug, moving through draft, review, then published; schedulable and fully versioned.
- **Navigation:** the site's header and footer link lists.
- **Copy:** every label that's content rather than code, per surface (site, portal, console, email).
- **Media:** the image library. Uploads go straight to private storage, alt text and focal points are editable, and public reads pass through a short-lived signed link so nothing is openly reachable.

Public content reads are edge-cached: fast for visitors, fresh within a minute of publishing.

## The Operations Center, page by page

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Every view keeps its state in the address bar (filters, tabs, open records), so links survive refresh and notifications deep-link to the exact task.

- **Workspace:** the day's triage. Unassigned consultations, cases awaiting a handler, invoices to raise, documents to review, reschedule requests, banded overdue / today / later, with inline assignment. A caseload view shows who is carrying what across the team. Alongside the derived work, staff keep their own task list: follow-ups with due dates, pinned to a lead or a case or neither, each reminding its owner exactly once when it falls due.
- **Inbox, Dashboard, Now:** what happened, the numbers, what's live.
- **Helpdesk:** the client conversation queue with its full lifecycle.
- **Chat:** staff messaging.
- **Consultations:** booked sessions. Assign, refer to another branch, delegate and reclaim, assess, reschedule approvals, no-show and credit, comments, document requests, and the full activity history.
- **Cases:** every engagement, list or board, with the full case file: stages, chapter owners, package and plan, invoices, comments, documents, visa and departure details, release override, post-arrival schedule, and referral to another branch.
- **Clients, Leads:** records and the enquiry pipeline.
- **Appointments, Live meetings:** the week and rooms in progress.
- **Universities, Programmes, Scholarships, Packages, Departure checklist:** the catalogue editors. Scholarships can also be awarded to a specific applicant from their record.
- **Invoices, Ledger, Payments, Fee schedule, Payment plans:** the finance suite.
- **Reports:** revenue and operations analytics.
- **Scheduling, My availability:** the week the branch offers, and each person's own hours and calendar connections.
- **Marketing:** campaigns, audiences, contacts, templates, automations.
- **System, Staff & roles, Clients directory, Authentication, Audit, Content (CMS), Lookups, Notifications, Settings:** platform administration. Health, the permission matrix, sign-in policy, the audit feed, content, form dropdowns, notification templates and integrations.

# The website and portal, page by page

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## THE PUBLIC WEBSITE

Open to everyone, no sign-in:

- **Marketing pages:** home, about, why choose us, services, visa services, student services, success stories ("Red Seat"), FAQs, blog, events, contact.
- **The catalogue:** destinations, universities, programmes and scholarships, each browsable down to a detail page and served live from the same database the staff edit.
- **The enquiry widget:** a floating door on every page where a visitor leaves their details and becomes a lead in the CRM.
- **The newsletter:** a subscribe prompt, double opt-in confirmation, and unsubscribe/preferences pages.
- **"Start your journey":** the doorway that turns a visitor into a registered client and drops them into chapter one.
- **The AI assistant and a newsletter popup** float on every public page; first-time assistant use passes a bot check.

## THE PORTAL

The sidebar is the journey itself, the six chapters as the spine, showing done, current and locked.

- **Home / Journey:** the spine, the current step, the next unlock.
- **Consultation:** choose online or in-person, pick a live slot, pay, join the call in-app, reschedule or cancel within policy.
- **Enrolment:** the assessment outcome, the decision, package and plan, the deposit.
- **Applications:** school choices, the fee invoice, per-school tracking, accepting an offer.
- **Visa:** consent, the invoice, live tracking to the decision (a refusal parks the case with a reapplication path).
- **Departure:** travel help (yes, hold or no), the fee milestone, the checklist, document release. Choosing travel help assigns a travel officer, and only then is the airline fare invoiced, separately: the service fee was already collected in the package, and no ticket is ever billed before someone owns the booking.
- **Complete:** the post-arrival instalment schedule.
- **Appointments, Documents, Fees, Security:** bookings, the vault, the ledger and receipts, and the security page: profile fields, a two-step verified email change, MFA setup, session list, and avatar upload with crop.
- **The Communication Center:** a floating support channel inside the portal, the staff thread plus the AI assistant.
- **Newsletter pages:** confirm, unsubscribe, and preferences.

# Data, security and the background engine

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## DATA POSTURE

Every table is behind row-level security. The app connects as owner, but any future non-owner path is denied by default rather than silently open. Anything that's evidence is append-only: audit entries, invoice events, booking events, assignment history, activity feeds. Corrections append; history is never edited.

## SECURITY POSTURE

**Identity always comes from the session, never from the request.** A caller cannot assert a role, branch or account id in a request body; the API derives all of it from the signed session cookie and fresh database reads on every request. A suspended account is refused on every request with its suspension reason. A staff member is just a user who also has a staff record linking them to a role and branch: staff are invite-only, there is no staff self-registration anywhere.

**The sign-in doors are rate-limited per address.** Sign-in attempts are capped at ten a minute, sign-ups at five, one-time-code sends and password-reset requests at three. The counters live in Redis and are keyed on the caller's address; if Redis is ever unreachable the limiter waits two seconds then lets the request through, so a cache outage can never lock everyone out.

**Credentials have real rules.** Staff passwords require twelve characters. One-time codes are six digits and die after ten minutes. Email-verification links, staff invitations and signed file links all expire on their own. Sessions expire too, and every account, client or staff, can list its active sessions and revoke them one by one; staff can also suspend a client account or revoke all of a client's sessions at once.

**Second factors are enforced per session.** A Google sign-in mints the session with the second factor still owed, and the API holds that session at a challenge until it clears. Email-code factors work the same way, and the code window dies with the session, so every new sign-in asks again. Whether a role must enrol is decided by role policy on the server, and the check runs in middleware before any handler.

**Sign-in methods are per-audience switches.** Password sign-in can be switched off for staff or for clients independently, and Google sign-in can be switched per audience too, so the attack surface each door exposes is a settings decision, not a rebuild.

**Secrets are encrypted at rest.** The OAuth refresh tokens behind staff calendar connections are stored under AES-256-GCM: tamper-evident, versioned ciphertext, so a leaked database dump does not leak anyone's calendar. A row that can no longer be decrypted, say after a key rotation, fails to a reconnect prompt rather than crashing availability lookups.

**The edge is strict.** Every response carries hardened security headers. The origin allowlist is exact-match and shared between CORS and the auth layer, so a sign-in callback can only ever point at a configured origin; localhost origins exist only outside production, and a malformed entry is rejected loudly at startup. Adding a domain is an environment variable, not a code change. Browser traffic reaches the API through each app's own proxy path, so session cookies stay first-party.

**Bots are checked at the edge.** The public enquiry form and the AI assistant sit behind a Cloudflare Turnstile check; a visitor who passes gets a signed verification cookie that the chat endpoint trusts, all stateless, nothing to store. Newsletter unsubscribe and preferences links carry a signature of the email address under the auth secret, so a preferences link for one address can never be rewritten to manage another.

**Webhooks authenticate by signature.** The Paystack webhook verifies an HMAC-SHA512 of the raw request body, the signature is the authentication, and a mismatch is refused outright. The other inbound webhook verifies its provider's signature headers the same way. Settlement itself is idempotent on the gateway reference, so a replayed or duplicated webhook settles once and no-ops after.

## DELETION, EXPIRY AND RETENTION

Different kinds of data leave the system in different, deliberate ways.

**Removing a client account** is a super-admin action, always audited, with three modes:

- **Disconnect** removes the login only. The applicant record, the cases and the files stay; the person simply can't sign in anymore.
- **Archive** marks the applicant record archived and removes the login. The history is kept but the person is closed out.
- **Purge** is the full removal: the person's conversations, comments, activity events and leads go, the applicant record and its cases are deleted, and every file they uploaded is cleaned from storage best-effort afterwards. One deliberate exception: **invoices are never deleted**. Journey invoices belonging to the purged case are detached and re-labelled as one-off charges, because financial records must outlive the case they came from.

**Everyday removal** follows the same spirit. Staff can remove leads, documents, canned replies, roles, packages, and marketing entities (draft campaigns, lists, templates, segments, individual suppressions, list members). Clients can withdraw a reschedule request, cancel autopay consent, remove a school choice, disconnect a calendar feed, or unsubscribe a device from push. Chat messages are never truly deleted: they're marked deleted and shown as a tombstone, because replies and forwards quoting them must not dangle. Open invitations can be revoked.

**Things that expire on their own:** staff invitations, one-time codes and verification links, sessions, signed file links (minutes for an upload, under an hour for a read), temporary consultation delegations, and a meeting's join window.

**Things kept on a clock:** rejected documents are purged by a daily sweep after a retention window; finished queue jobs are kept only briefly. **Things kept forever:** the audit trail, invoice events, booking events, assignment history and activity feeds are append-only and are never purged by anything.

## BACKGROUND JOBS

Work that shouldn't block a request runs on a separate worker process: emails, campaign sends, push notifications, instalment charges, offline chat replies, helpdesk SLA checks, automation date-triggers, meeting status polls, calendar feed syncs, and document cleanup. Until that process runs, jobs simply wait. Bookings still succeed, but no email sends. It is not optional in production.

## EXTERNAL SERVICES

Paystack for all payments, Resend for all email (delivery, open, bounce and complaint events flow back through a signed webhook), LiveKit for video rooms (with Daily kept as an alternative provider), Supabase for private file storage, Google for client sign-in and staff calendars, web push for browser notifications, and a pluggable text provider for phone sign-in (unconfigured by default; the feature refuses rather than pretending).

## CONFIGURATION AND DEPLOYMENT

Production refuses to boot without its database, queue, signing secret and the browser-facing addresses. Misconfiguration fails loudly at startup, not silently at runtime. The API ships as a Docker image; the front ends deploy as Cloudflare Workers. The database is migrated by numbered files; new databases are built directly from the schema because the historical migration chain doesn't replay cleanly from zero. The rules and the reasons are written down in the migration discipline notes.

## Under the hood, the flows that matter

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Features above; mechanics here. These are the paths worth understanding before changing anything.

### EVERY REQUEST, IN ORDER

A call reaches the API, gets security headers, passes the origin check, passes rate limiting, then meets the guard chain in a fixed order: prove you're signed in, prove you're staff (if the route needs it), prove you've enrolled a second factor (staff), prove your role may see the module, prove your role may do the action. A branch-scoped role gets its branch constraint applied last. The permission check reads the live roles table, so a role edit takes effect without a deploy; the built-in matrix is only a fallback. Inputs are validated by the same schema that generates the public API reference, so docs and enforcement can't disagree.

### HOW THE JOURNEY STEP IS DECIDED

Nothing stores "step 7 of 14". Each time the portal asks, the API collects the facts about the client's newest case: does a consultation exist and what was its outcome, what did the client consent to, which package and plan, which invoices are paid, which handler is assigned, which schools are locked, which offers arrived, what the visa record says, what travel help was chosen, what checklist items are done. A single shared function turns those facts into the step list. Because it's computed fresh, the two apps can never disagree, and a fact that appears out of order (a visa paid before schools locked) marks the earlier steps "skipped" rather than pretending they were done.

### HOW A BOOKING BECOMES A CONSULTATION

The client picks a slot, the API creates a payment, Paystack collects the money, and only then do the records land: the booking, the consultation, the invoice marked paid, the event history, and queued notifications. The double-booking guard is a database rule, not application hope: the column ranges physically cannot overlap, so two simultaneous checkouts can't both win the slot. Everything that can fail after the commit (the email, the meeting link, the push) is queued, so a provider outage can never roll back a paid booking.

### HOW A PAYMENT SETTLES, TWICE, SAFELY

Two paths can settle a payment. The fast path is the client returning from checkout and the portal asking the API to verify. The authority path is the signed webhook arriving from Paystack on its own schedule. Both run the same settlement, keyed on the gateway reference: the first one to arrive writes the payment, the ledger entry, the event history, and the journey side effects (a deposit fires the handler handoff exactly once). The second arrival sees the reference already recorded and becomes a no-op. Retries, replays and double submissions cannot charge or advance twice. If neither lands, staff reconcile against the gateway to close the gap.

## HOW INVOICES ARE NUMBERED AND TRACKED

Drafts (proformas) and issued invoices draw from separate number sequences, each allocated inside the same transaction as the insert, so there are no duplicate numbers under concurrency and no gaps from abandoned drafts. Paid totals are never stored as a number that can drift; they're summed from the payment records each time, and the next uncovered instalment due date is derived from how much has been covered. The ledger views are projections over the invoice event history, which is why they can't disagree with the payments table.

## HOW A DOCUMENT MOVES

Upload is three steps: the API mints a short-lived upload link, the file goes straight from the browser to private storage, then the client tells the API it's complete and the record lands. Review changes the record's state (verified, or rejected with a note the client sees) and that state feeds the checklist and the journey. Downloads are minted per request. Rejected files are deleted by a daily sweep after a retention window, so rejected personal documents don't accumulate in storage.

## HOW MARKETING ACTUALLY SENDS

When a campaign is sent, the audience is frozen into a recipient ledger: opted-in and unsuppressed people become pending rows, everyone else becomes a skipped row with its reason recorded. The worker then walks pending rows only, which is why a retry after a crash can never re-mail someone already delivered. Per recipient it builds the merge context from the real case (officer name, branch, next due), wraps every link for click tracking, sends through the provider, and stamps the outcome plus the provider's message id back on the row. As delivery events return through the signed webhook (open, bounce, complaint), the recipient row is updated and a bounce or complaint also writes the global suppression, so the address is protected everywhere from then on. The report is a read of the ledger, so numbers can never drift from what actually sent.

Automations work the opposite direction: a domain moment (visa approved, offer received, assessment complete, no-show, milestone overdue) fires an event into an intake check, which writes a scheduled send per matching person. A sweep every minute delivers whatever's due; a daily scan inside the same sweep handles date triggers like thirty-days-to-departure. Every firing is logged, including the skip reason.

## HOW CONSENT FOLLOWS AN ADDRESS

Consent lives on the address, not on a list row. A confirmation click, an offline-consent note, a re-opt-in on the preferences page, an unsubscribe, a bounce: all of it lands in one consent ledger keyed by address. Any surface that asks "may we email this person" reads that ledger plus the global suppression list, so consent granted in the portal and a suppression caused by a bounce protect every list and every campaign at once.

## HOW STAFF PERMISSION RESOLVES

A staff sign-in produces the same session as a client sign-in; what makes it staff is the staff record attached to the account, carrying role and branch. When a guarded route runs, the role's permission list is read from the roles table (a custom role's list is exactly what the permission matrix saved), falling back to the built-in defaults only when the role has no row. Seeing and doing are separate entries in that same list: "invoices" the module opens the page, "issue invoices" the capability enables the action. Ownership of a chapter is a capability too, which is why the visa-officer picker can only offer staff who may own visa work.

## HOW CHAT STAYS LIVE

Each participant holds a read cursor on a conversation; unread counts are the distance between that cursor and the latest message. Typing indicators and staff presence ride on lightweight heartbeats rather than sockets. When a client is offline, a reply is emailed; when they answer the email, the reply is threaded back onto the same conversation by the token in the subject, so a thread started in the portal can continue in an inbox without forking.

## THE AUDIT CHAIN, MECHANICALLY

Every audited write computes a fingerprint over the entry's content plus the previous entry's fingerprint, and stores both. Verification replays the whole chain: if any entry was edited or deleted, every fingerprint after it stops matching. It proves tampering happened, not just that data changed, which is why the console shows the chain status as a badge rather than trusting the rows.

## THE SWEEPS AND THEIR CLOCKS

- Every minute: due automation sends are delivered; inside the same run, a daily pass handles date triggers.
- Every fifteen minutes: unclaimed helpdesk requests are auto-assigned so nothing sits unanswered.
- Every sixty seconds: meetings are marked live or ended.
- Daily: rejected documents are purged; date-based automation triggers evaluated.
- On a schedule: staff calendar feeds are mirrored into busy blocks.

Every queued job carries an idempotency key; a duplicate key is a silent no-op, failed jobs retry with growing backoff, and finished jobs are kept only briefly, so the queue can't fill with history.

## HOW THE RECORDS RELATE

One account ( `users` ) may be a client, a staff member, or both shapes at once (a staff record attaches to the account). A client has a profile; each engagement is a case; a case owns school submissions, invoices, documents, comments, chapter seats and handoffs. An invoice owns lines, payments and events; payments point at gateway transactions; stored authorizations power instalment charges. A conversation owns members, messages, attachments and reactions. A campaign owns a recipient ledger and tracked links; the

consent ledger and suppression list stand beside all of it, keyed by address rather than by list. People in marketing are assembled at read time from three sources (list contacts, applicants, leads), which is why one person can never appear as two rows.

*Covers the three deployables (public site + portal, Operations Center, API + worker) and the shared packages, as of the current codebase.*